

Computer Networks

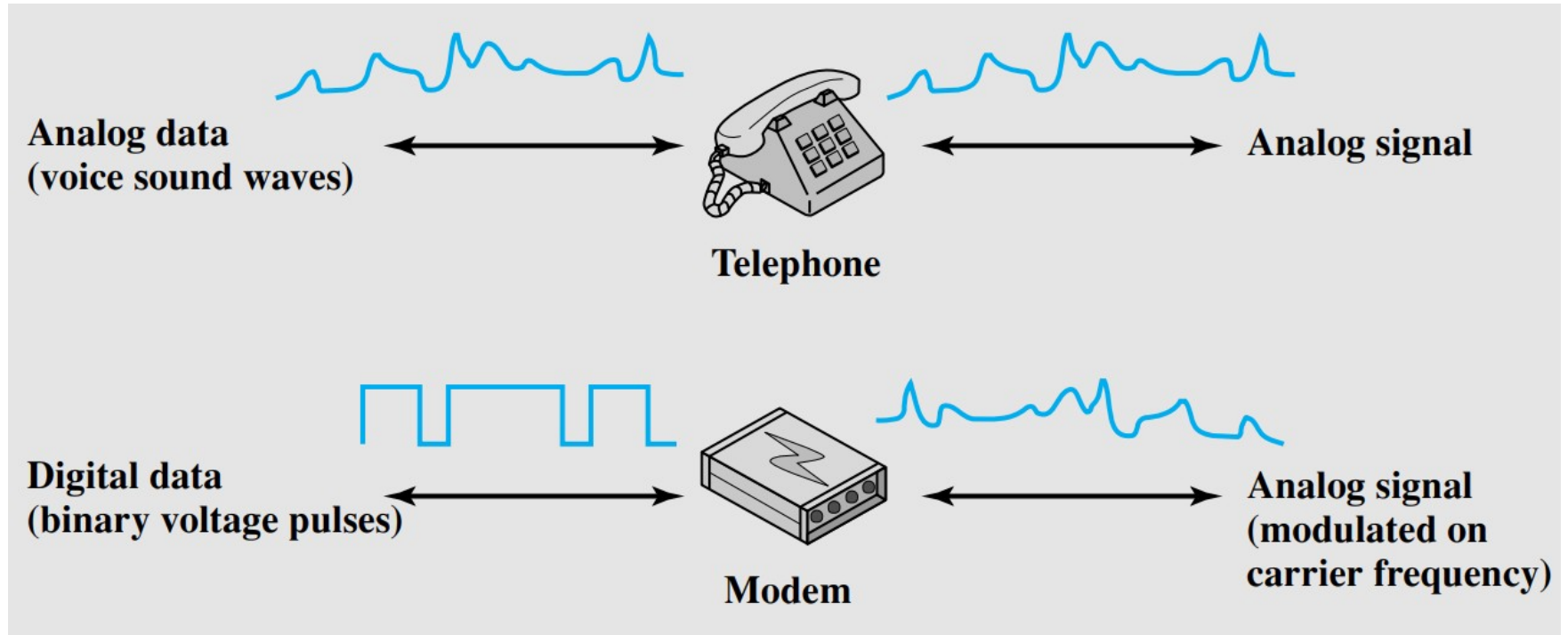
Signal Encoding Techniques (Digital to Digital)

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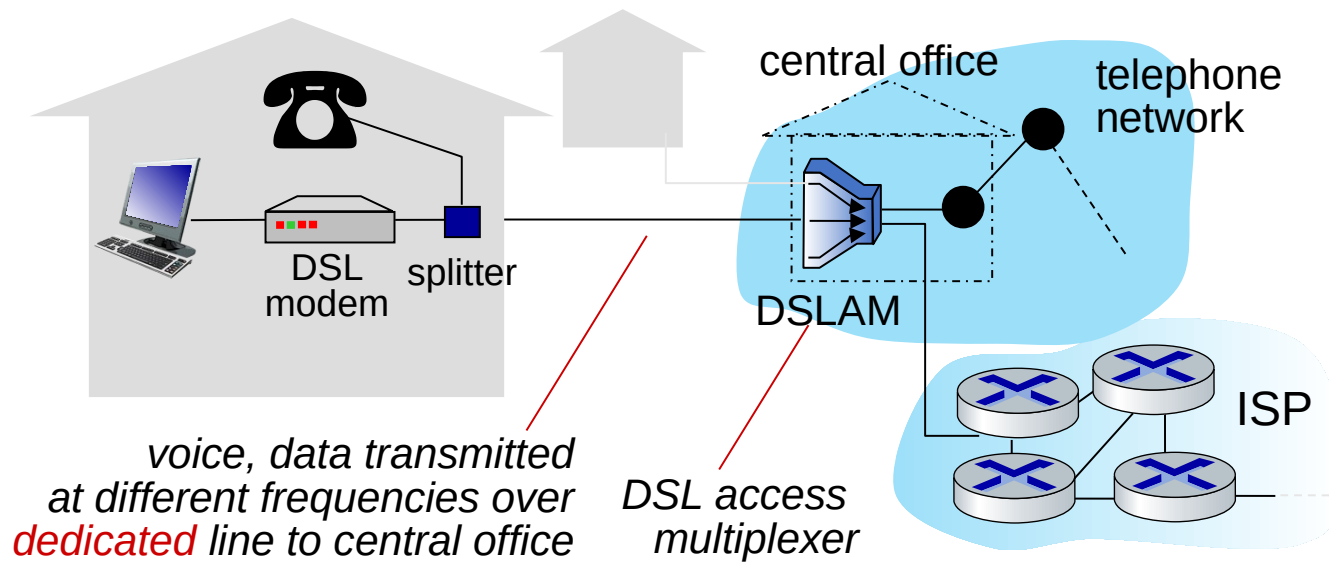
Computer Science and Engineering

IIT Kanpur

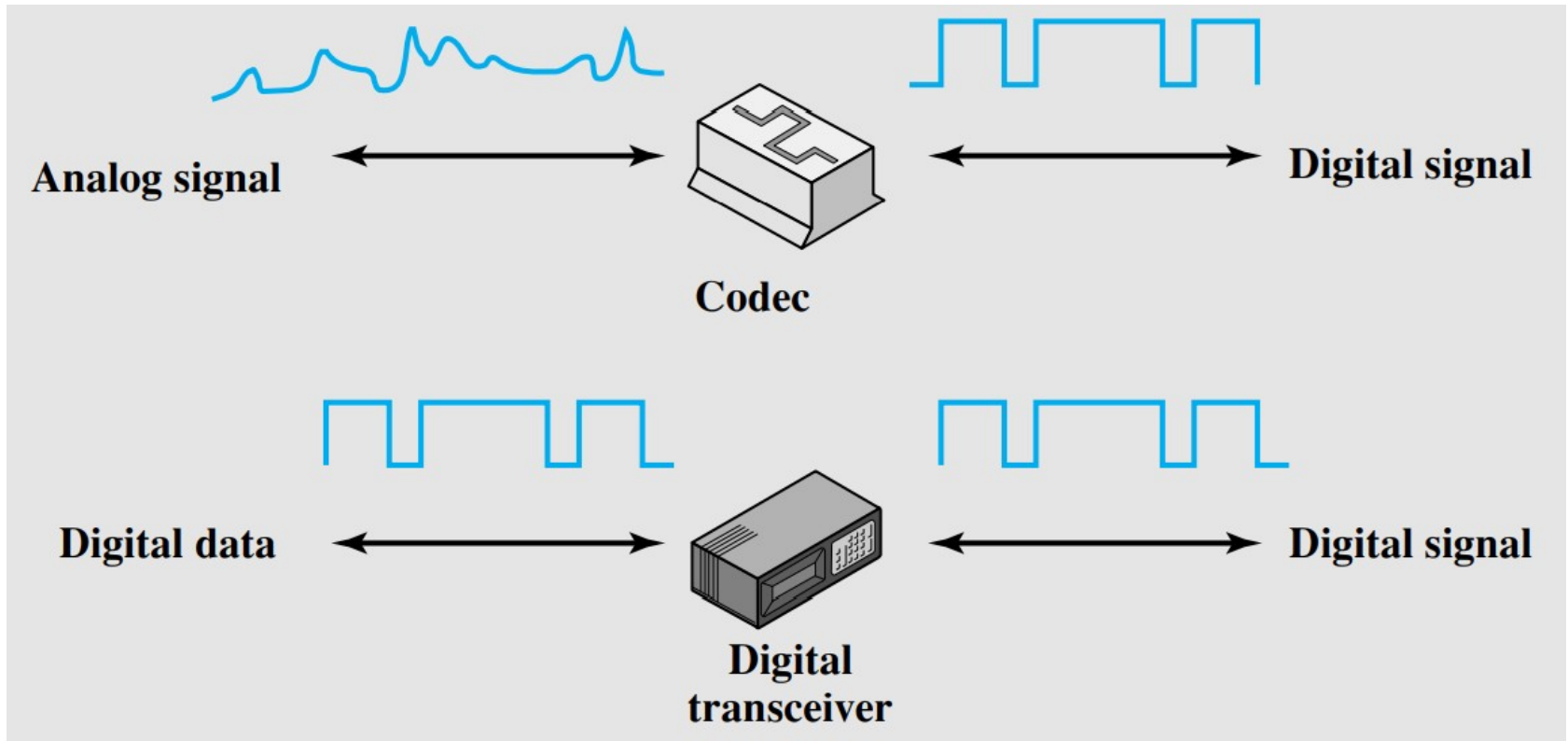
Analog Signal



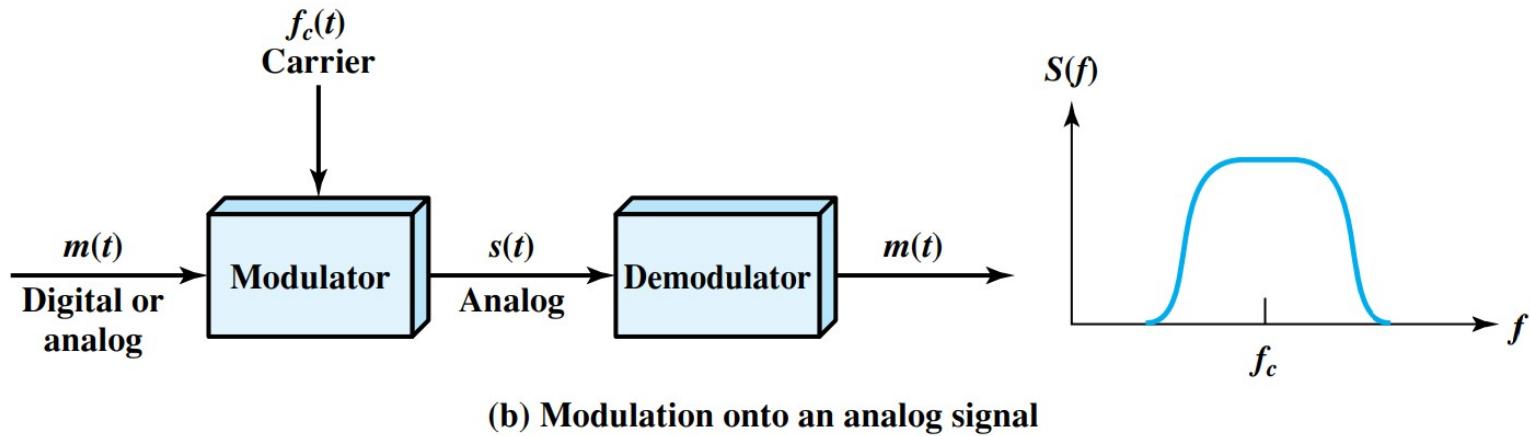
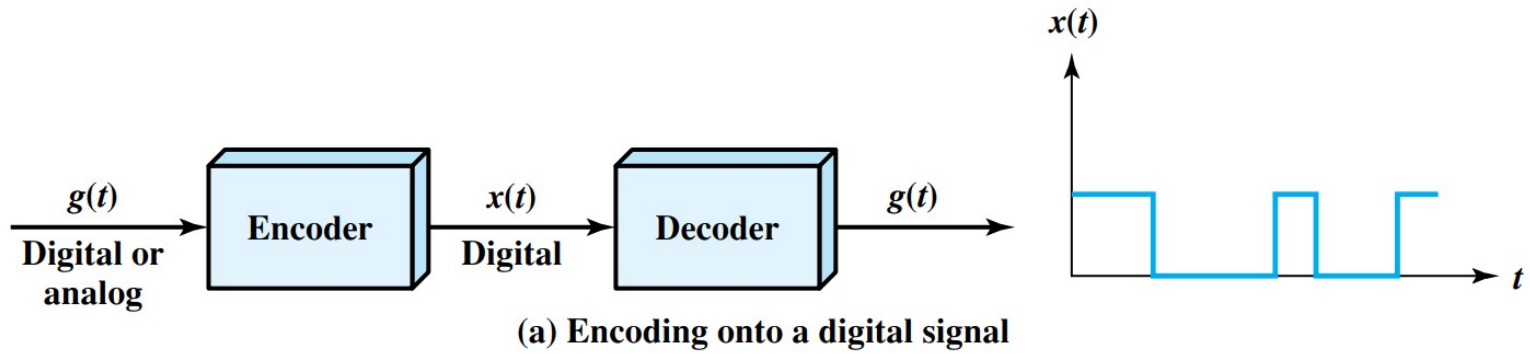
Analog Signal



Digital Signal

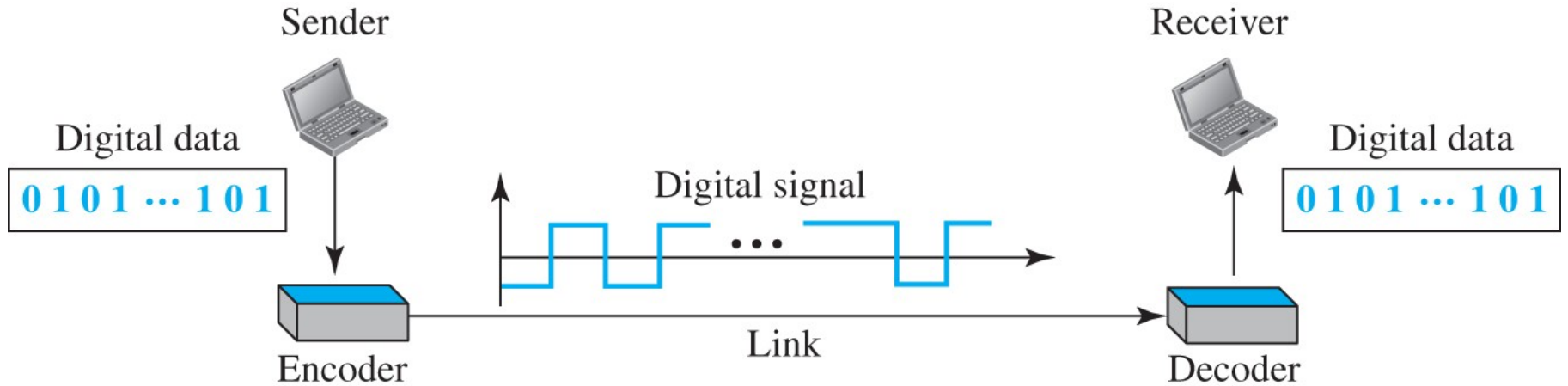


Signal Encoding Techniques



Digital Data → Digital
Signals

Digital Data → Digital Signals



Evaluation Criterias of Encoding Techniques

❑ Signal spectrum, DC component and bandwidth

- ❑ No DC component is desirable
- ❑ Encoding with less bandwidth is preferable

❑ Clocking

- ❑ Transmitted signal can be used by the receiver for synchronizing

❑ Error detection

- ❑ Encoding scheme can have some inbuilt error detection mechanism
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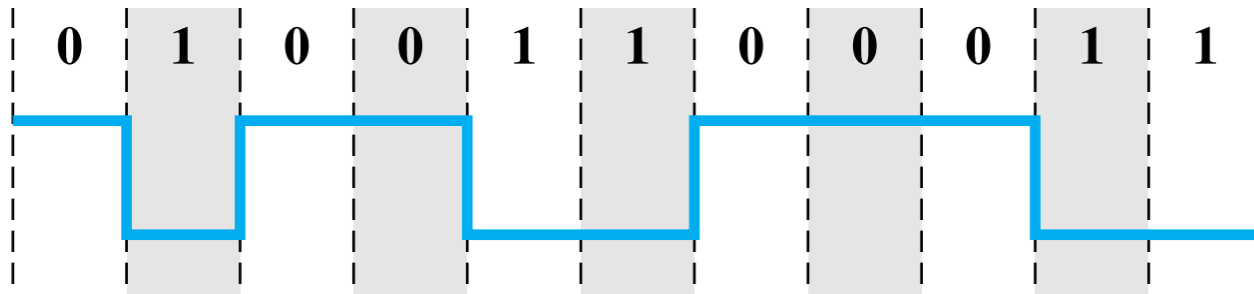
NRZ-L

$$\text{PSD}(f) = A^2 T_b \text{sinc}^2(\pi f T_b)$$

0 = High level

1 = Low level

- Problems:
 - Lack of clock recovery during long string of 0 or 1s $\square$
 - Has d.c. component $\square$



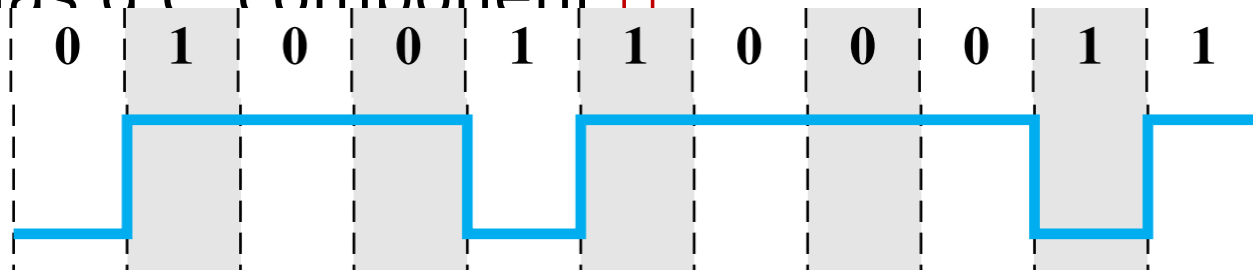
NRZ-I

$$\text{PSD}(f) = A^2 T_b \text{sinc}^2(\pi f T_b)$$

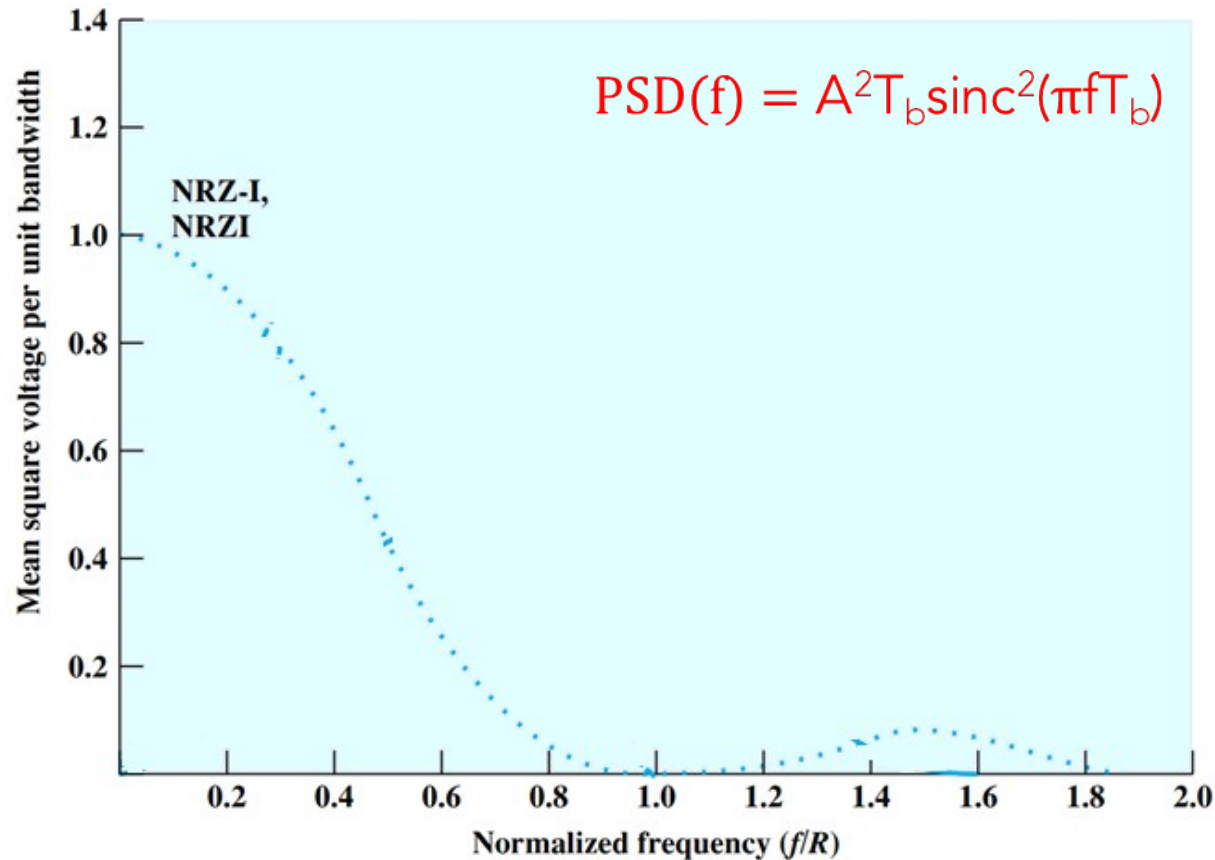
1 = change of signal level (on-off or off-on)

0 = no change of signal level

- Fixes clocking problem for long string of 1 bits □
- Problems:
 - Lack of clock recovery during long string of 0 bits □
 - Has d.c. component □



Performance of Encoding Schemes



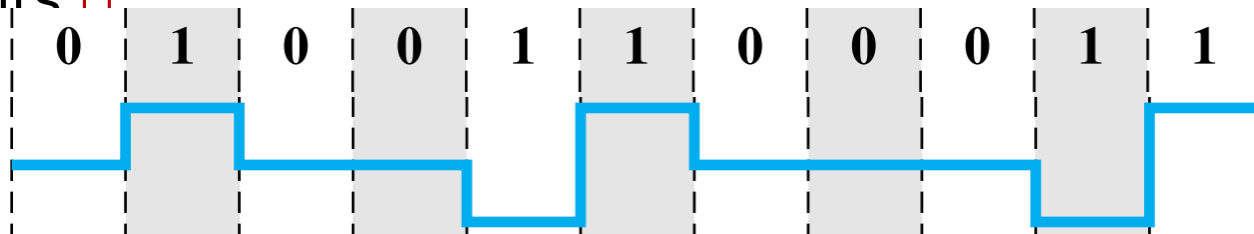
Bipolar-AMI

Uses 3 signal levels: +V, 0, -V $\text{PSD}(f) = A^2 T_b \text{sinc}^2(\pi f T_b) \sin^2(\pi f T_b)$

0 = no signal (0 voltage)

1 = alternating +V and -V

- No d.c. component (alternating +V and -V) □
- Can detect some bit errors (consecutive +V or -V) □
- Problems:
 - Loss of synchronization during long string of 0 bits □
 - Inefficient usage: 3 signal levels is used to represent 2 bits □



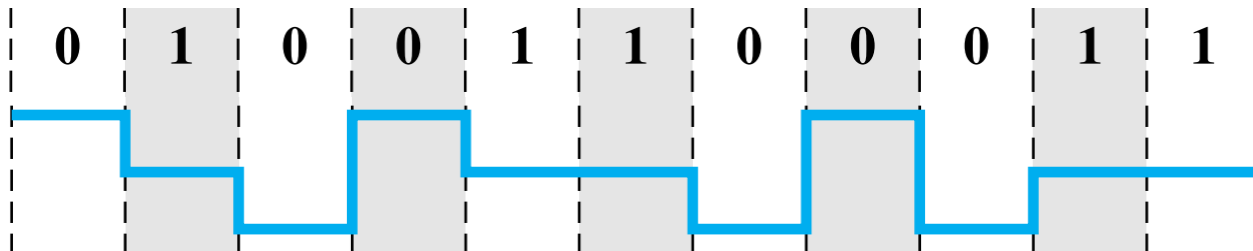
Pseudoternary

$$\text{PSD}(f) = A^2 T_b \text{sinc}^2(\pi f T_b) \sin^2(\pi f T_b)$$

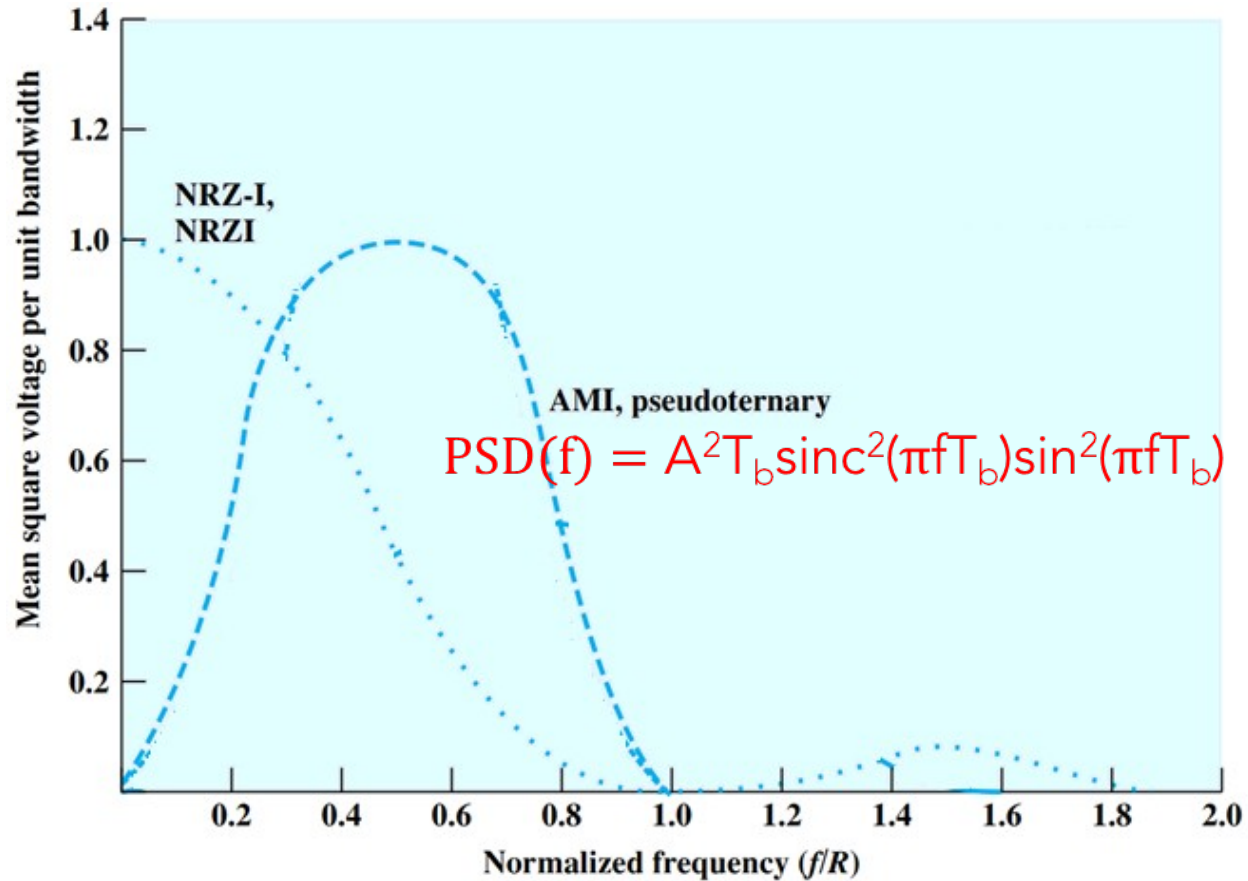
Same as Bipolar-AMI except reverses signaling:

1 = no signal (0 voltage)

0 = alternating +V and -V



Performance of Encoding Schemes



Manchester Encoding

Always transition in middle of bit period:

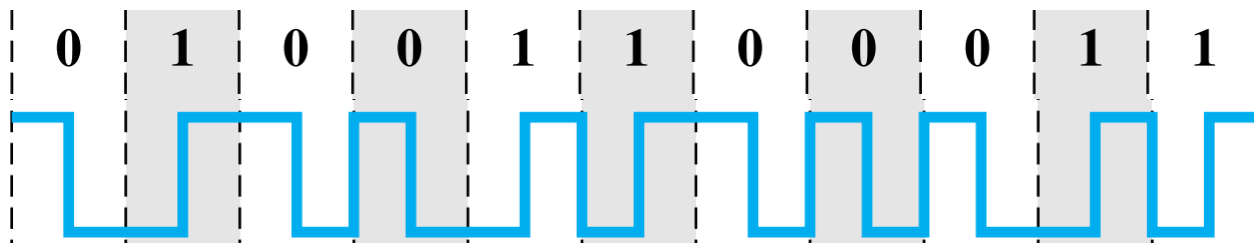
0 = high-to-low transition

1 = low-to-high transition

$$\text{PSD}(f) = A^2 T_b \text{sinc}^2 \left(\frac{\pi f T_b}{2} \right) \sin^2 \left(\frac{\pi f T_b}{2} \right)$$

Transition at **beginning** of bit period when necessary

- Good clock recovery, error detection capability, no d.c. component □
- Bandwidth requirement is higher □



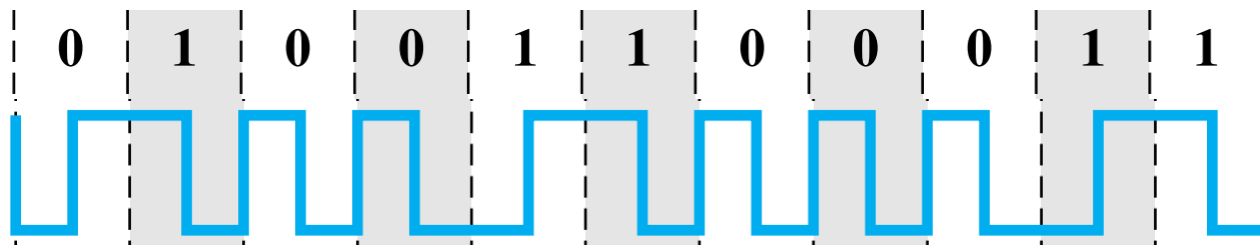
Differential Manchester Encoding

$$\text{PSD}(f) = A^2 T_b \text{sinc}^2\left(\frac{\pi f T_b}{2}\right) \sin^2\left(\frac{\pi f T_b}{2}\right)$$

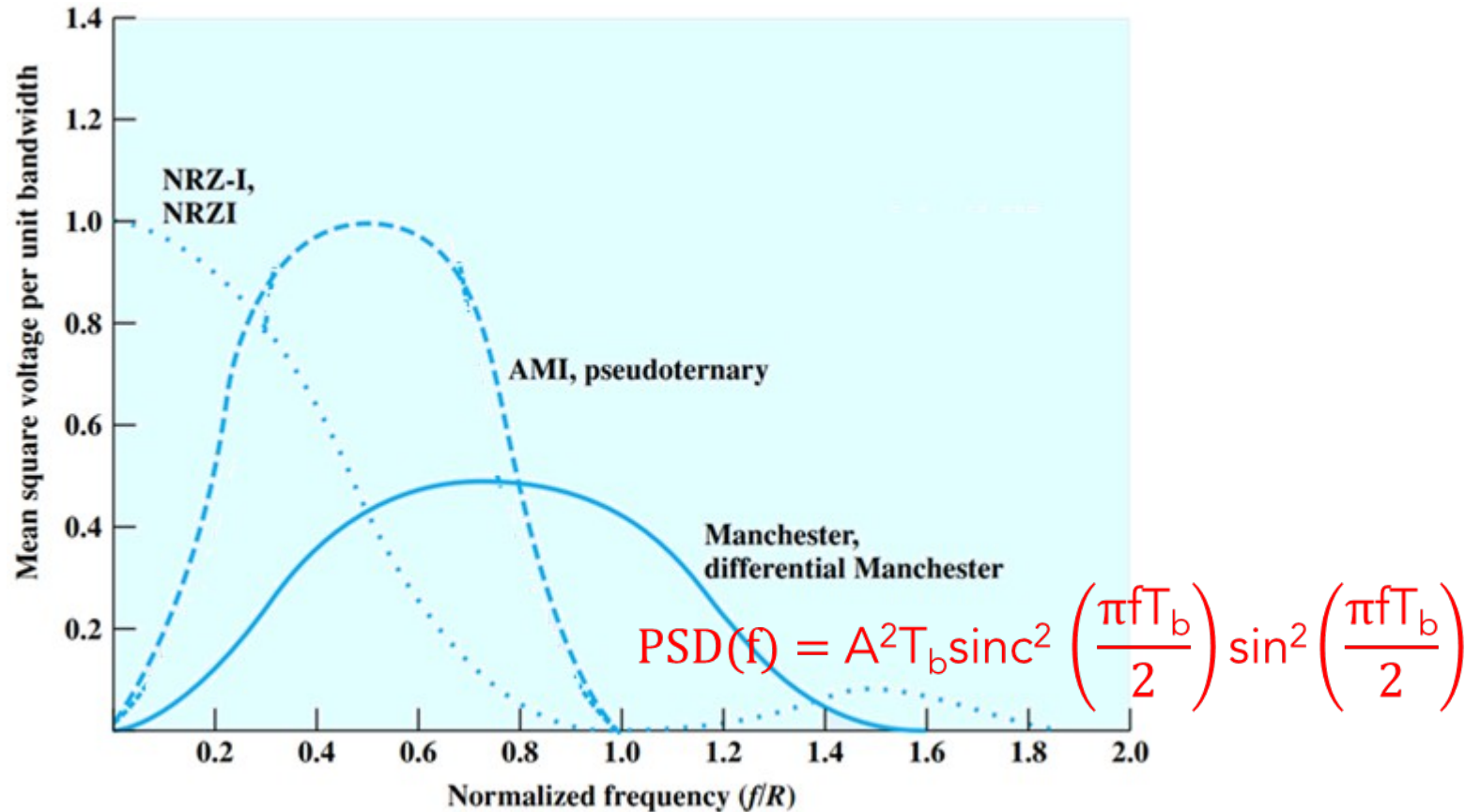
0 = transition at beginning of bit period (low-to-high or high-to-low, depending on previous output level)

1 = no transition at beginning of bit period

- Similar properties as Manchester encoding
- Bandwidth requirement is higher $\square$



Performance of Encoding Schemes



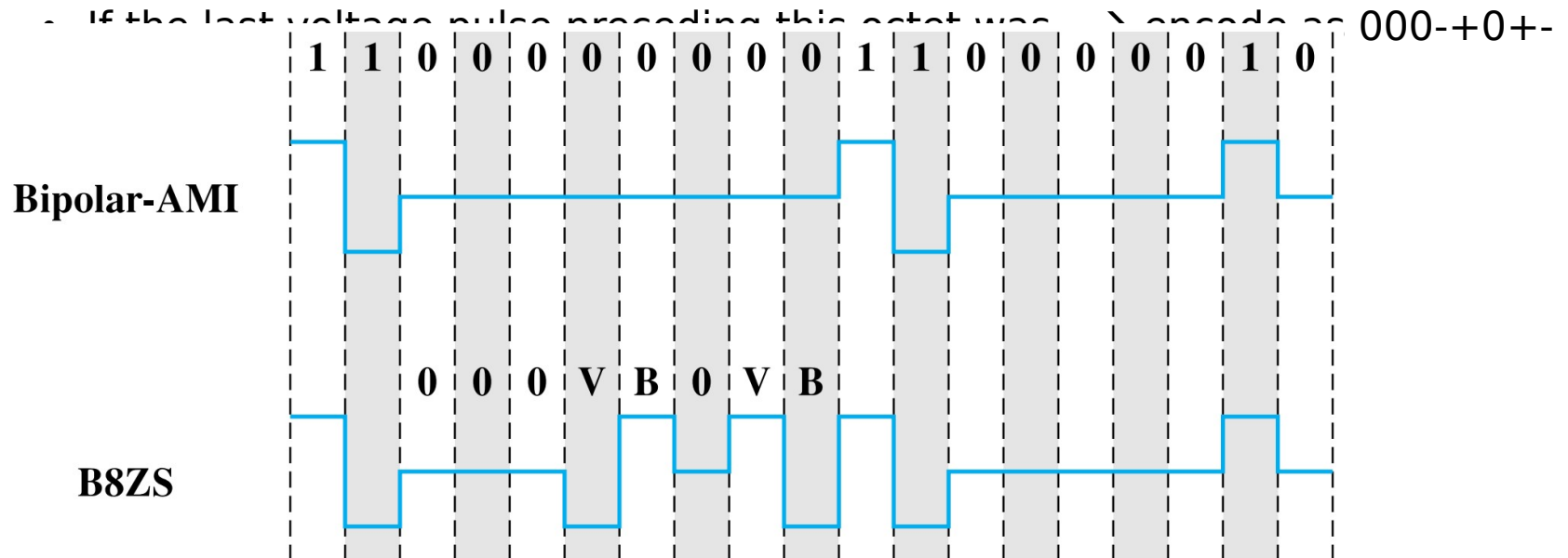
B8ZS

❑ Based on **Bipolar-AMI**

❑ 8 consecutive 0s are encoded as:

- If the last voltage pulse preceding this octet was + → encode as 000+-0-+

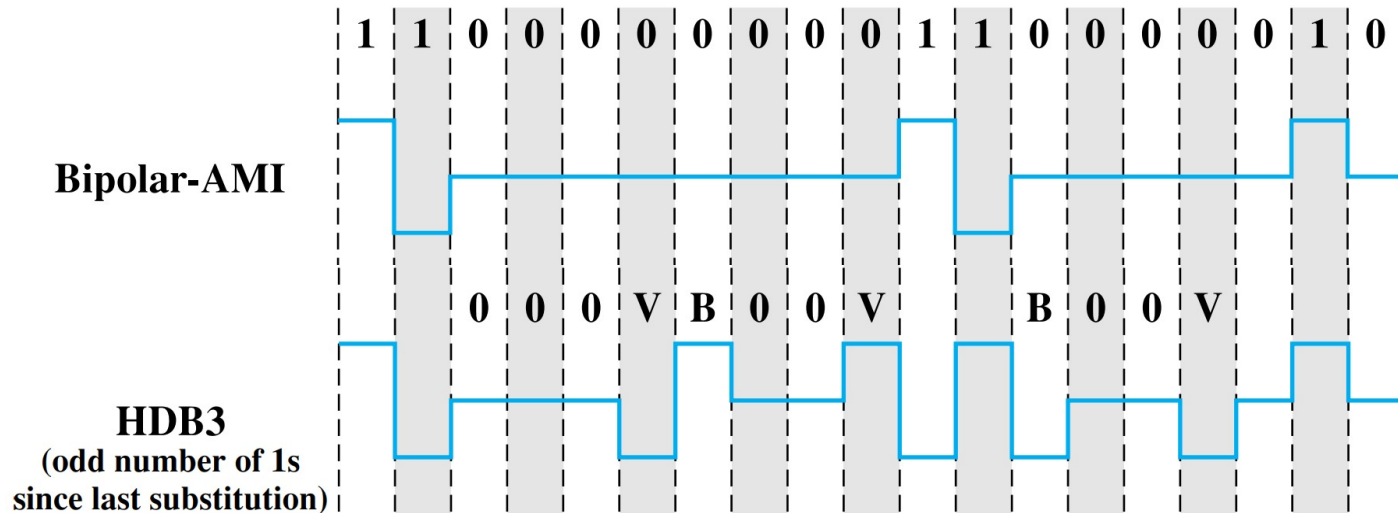
• If the last voltage pulse preceding this octet was - → encode as 000-+0+-



HDB3

❑ Based on **Bipolar-AMI**

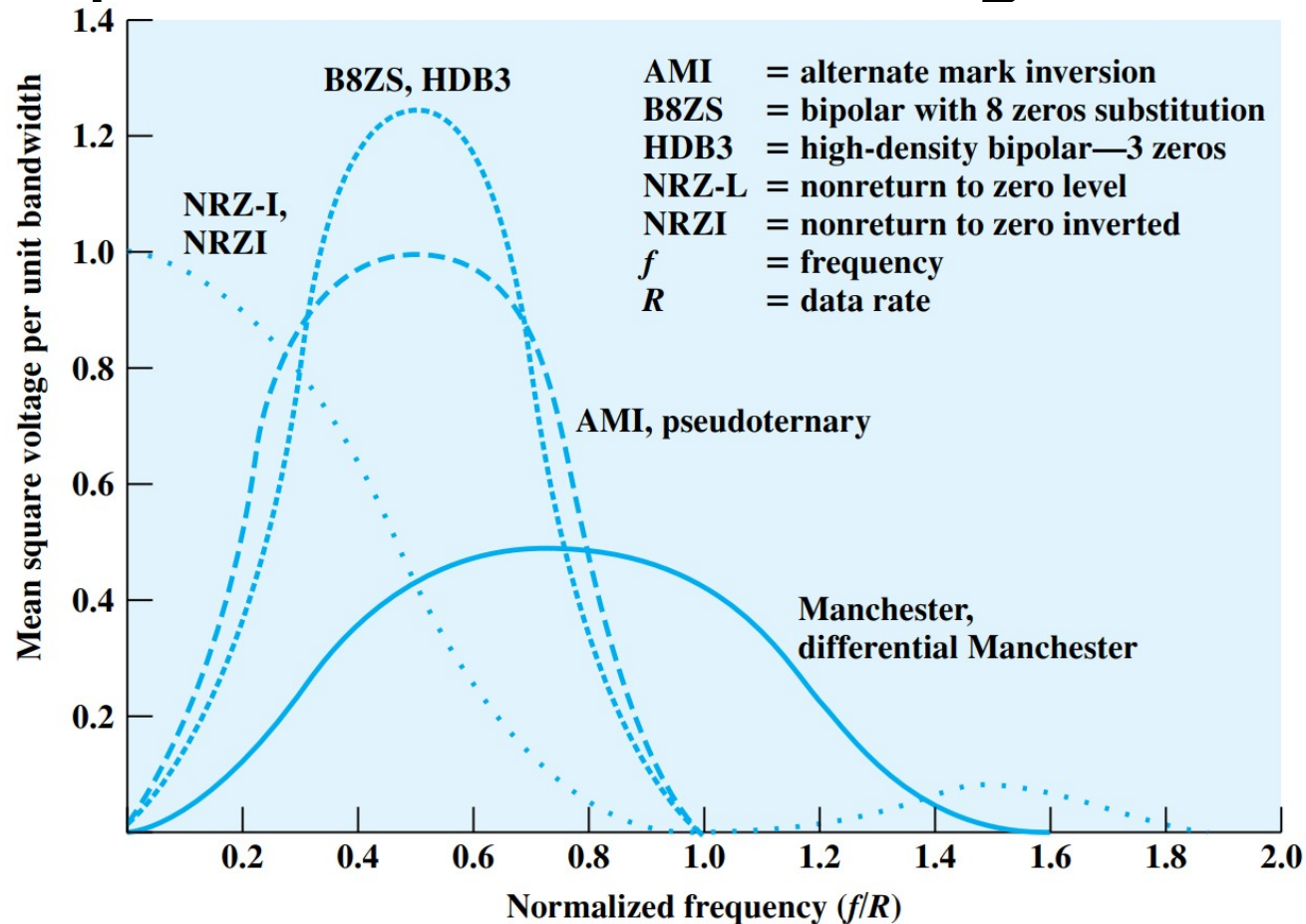
Polarity of Preceding Pulse	Number of Bipolar Pulses (ones) since Last Substitution	
	Odd	Even
–	0 0 0 –	+ 0 0 +
+	0 0 0 +	– 0 0 –



Advantages B8ZS and HDB3

- No d.c. component □
 - Fixes clocking problem for long string of both 0 and 1s □
 - Error detection capability □
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Comparison of Encoding Schemes



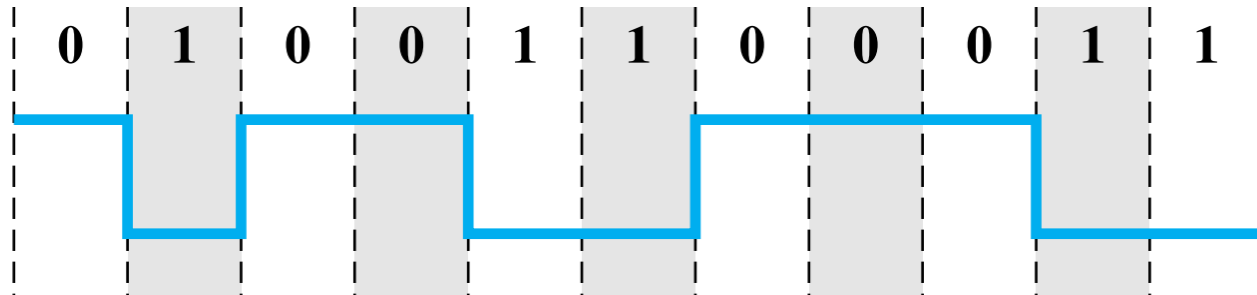
Unipolar vs Polar Encoding

❑ Polar

- One logic state represented by positive voltage and the other by negative voltage

❑ Unipolar

- All signal elements have the same sign



Summary

□ Signal encoding techniques (Digital data → Digital signals):

- Different encoding techniques discussed
 - NRZ-L and NRZ-I
 - Bipolar-AMI and Pseudoternary
 - Manchester and Differential Manchester
 - B8ZS and HDB3
 - Comparison of the encoding schemes
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